

you save the file, put a '.m' extension on it. This lets FreeMat know that it is a usable script.

The permutation function is similar to the combination function. And it is even simpler. The code for it is as follows:

```
function return_value=perm(n,r)
return_value=gamma(n+1)/gamma(n-r+1);
```

Just as we did earlier, save the file to a directory within the working path.

Test your functions

Let us do a couple of simple tests on your new functions:

```
--> comb(7,3)
ans =
35
--> perm(7,3)
ans =
210
```

If you get an error

The most common errors when trying to use functions is:

"Error: Undefined function or vari-

Variables in functions

Parameters passed to the function (parameter1, parameter2, etc) can have descriptive variable names. In the case of functions for combination and permutation below, we'll use variables 'n' and 'r'. The reason is that these are the common names for variables when they are described in textbooks and references.

Variables used in functions are separate from those within the script or program that called them. For example, in our function below, we'll use variables 'n' and 'r'. However, if we use these variables within FreeMat, it will not cause a problem. FreeMat considers them separate. Clear as mud? Don't worry if it does not make sense. Just remember: Variables used within FreeMat and those within functions can have exactly the same names and FreeMat won't get confused.

Our function for calculating combinations will use the function name of 'comb.' The code is as follows:\

```
function return_value=comb(n,r)
return_value=gamma(n+1)/(gamma(n-r+1)*gamma(r+1));
```

able"

This means that FreeMat cannot find the function. Double-check that you typed the function name correctly. Also check that the directory where the files are stored is in the path set by the path tool (Tools→Path Tool).

When you save the function, you need to give the file the same name as the function (except with '.m' appended). The file name must match what

you type to call the function exactly and is case-sensitive too; i.e., Comb(3,7) != comb(3,7). Check your naming conventions if function files are there but still things aren't working.

Software available on DVD

This tool is available on this month's DVD, for both Windows and Mac PCs. ●

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suggestion or a complaint?

You can e-mail it to
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of the rest.



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